

Performance Testing

Date	30 September 2026
Team ID	SWTID-2026-1307
Project Name	Script-Controlled ACL-Restrict Record Access Based on Field Value.
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	ServiceNow System Diagnostics, Postman for ACL API check
Type of Testing	Load Testing, Security / Access Control Testing
Target Module / API	Scripted ACL Module - x_hr_confidential_table with confidential field
Test Environment	Cloud / ServiceNow PDI (Personal Developer Instance)
Test Date	28-09-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Verify record access when confidential=false for regular user	10	60	Access Allowed - Record should be visible
2	Verify record access when confidential=true for regular employee	20	60	Access Denied - ACL should block record
3	Verify record access when confidential=true for HR Manager role	20	60	Access Allowed - Only HR Manager can view

4	Load test with 50 concurrent users querying confidential table	50	120Access Allowed - Record should be visible	Response time < 3 sec, ACL logic should not fail under load
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 seconds	Pass	ACL check optimized with Script Include
2	Response Time (Max)	< 5 seconds	2.8 seconds	Pass	Max time during 50 concurrent users
3	Throughput (Req/sec)	-	45 Req/sec	Pass	Stable under load test
4	Error Rate	< 1%	0.2%	Pass	No ACL bypass errors
5	CPU Utilization	< 80%	62%	Pass	Within limit during peak load
6	Memory Utilization	< 80%	58%	Pass	Efficient script execution

Step 4: Observations & Analysis

Key Findings:

The performance test results showed that the Scripted ACL module performs efficiently even under concurrent load. The average response time was 1.2 seconds, which is well below the target of 2 seconds. Security enforcement using field-based ACL did not cause any major delay. The system maintained stability with 50 virtual users and low error rates.

Bottlenecks Identified:

1. Slight increase in response time was observed during peak load of 50 concurrent users.
2. No major bottlenecks found, but complex Script Include logic can impact performance if not optimized

.Optimization Steps Taken:

Used a reusable Script Include instead of writing logic directly in ACL to improve modularity and reduce execution time. Added indexed field confidential in the table to speed up query filtering. Implemented proper role checks using `gs.hasRole()` to avoid redundant database calls.